

Fondamentaux théoriques du machine learning

Risks and risk decompositions

Classification and regression trees

Deterministic bound on the estimation error

We consider the best estimator in the hypothesis space F .

$$f_a = \arg \min_{h \in F} R(h)$$

Exercise 1: Let us show that

$$R(f_n) - R(f_a) \leq 2 \sup_{h \in F} |R(h) - R_n(h)| \quad (1)$$

Deterministic bound on the estimation error

$$f_a = \arg \min_{h \in F} R(h)$$

$$\begin{aligned} R(f_n) - R(f_a) &= (R(f_n) - R_n(f_n)) \\ &\quad + (R_n(f_n) - R_n(f_a)) \\ &\quad + (R_n(f_a) - R(f_a)) \\ &\leq |R(f_n) - R_n(f_n)| \\ &\quad + (R_n(f_n) - R_n(f_a)) \\ &\quad + |R_n(f_a) - R(f_a)| \\ &\leq 2 \sup_{h \in F} |R(h) - R_n(h)| \\ &\quad + (R_n(f_n) - R_n(f_a)) \end{aligned} \tag{2}$$

But by definition f_n minimizes R_n , so $(R_n(f_n) - R_n(f_a)) \leq 0$.

Risks and risk decompositions

Classification and regression trees

- ▶ Y_{pred} is the random variable representing this prediction (proportional)
- ▶ Y is the random variable representing the class, in this node (empirical distribution)

$$\begin{aligned} P(Y_{pred} \neq Y) &= \sum_{l=1}^L P(Y_{pred} \neq Y | Y = l) P(Y = l) \\ &= \sum_{l=1}^L \left(1 - P(Y_{pred} = Y | Y = l)\right) P(Y = l) \quad (3) \\ &= \sum_{l=1}^L (1 - p_n^l) p_n^l \end{aligned}$$

Homogeneity criterion for classification : Gini impurity

$$H(n) = \sum_{l=1}^L p_n^l (1 - p_n^l) \quad (4)$$

If we predict the classes in node n according to the proportions of the labels in n , then the Gini impurity is the probability of making a mistake, given that we are in node n .